

File: Shapes/Square.py

```
# Square class

import pygame
import random

# Set up the colors
RED = (255, 0, 0)
GREEN = (0, 255, 0)
BLUE = (0, 0, 255)

class Square():

    ❶ def __init__(self, window, maxWidth, maxHeight):
        self.window = window
        self.widthAndHeight = random.randrange(10, 100)
        self.color = random.choice((RED, GREEN, BLUE))
        self.x = random.randrange(1, maxWidth - 100)
        self.y = random.randrange(25, maxHeight - 100)
        self.rect = pygame.Rect(self.x, self.y, self.widthAndHeight,
                                self.widthAndHeight)
        self.shapeType = 'Square'

    ❷ def clickedInside(self, mousePoint):
        clicked = self.rect.collidepoint(mousePoint)
        return clicked

    ❸ def getType(self):
        return self.shapeType

    ❹ def getArea(self):
        theArea = self.widthAndHeight * self.widthAndHeight
        return theArea

    ❺ def draw(self):
        pygame.draw.rect(self.window, self.color,
                         (self.x, self.y, self.widthAndHeight,
                          self.widthAndHeight))
```

Listing 9-2: The square class